

CSCI 332, Fall 2026

Homework 3

Due Monday, September 14 Anywhere on Earth (6am Tuesday)

Submission Requirements

- Type or clearly hand-write your solutions into a PDF format so that they are legible and professional. Submit your PDF on Gradescope.
- Do not submit your first draft. Type or clearly re-write your solutions for your final submission. If your submission is not legible, we will ask you to resubmit.
- Use Gradescope to assign problems to the correct page(s) in your solution. If you do not do this correctly, we will ask you to resubmit.

Academic Integrity

Remember, you may access **any** resource in preparing your solution to the homework. However, you **must**

- write your solutions in your own words, and
- credit every resource you use (for example: “Bob Smith helped me on this problem. He took this course at UM in Fall 2020”; “I found a solution to a problem similar to this one in the lecture notes for a different course, found at this link: www.profzeno.com/agreatclass/lecture10”; “Here is a link to the full transcript of my chat with ChatGPT about the problems on this HW”.) Ask if you need any clarifications about how to properly cite your sources!

1. (0 points) List the resources you used for each problem in this homework. If you used no resources except lecture notes and videos and the linked textbook, you can say “none”. If you used an AI tool, you should put a link to your full conversation here, and you must use the [provided prompt](#) to guide the AI to act as a tutor, not an answer-provider.

You will be asked to resubmit your homework if this section is blank.

Remember that no matter what resources you consulted, you must write your solutions *in your own words*.

2. (3 points) Consider the number of the syllables in the pronunciation of the integer n . For example, when $n = 1$, the pronunciation is “one” and there is one syllable. For each of the following, give the tightest/simplest function possible, or state that there is no function possible.
 - (a) Give a function $f(n)$ so that the number of syllables in the pronunciation of n is $O(f(n))$ for all $n \geq 1$.
 - (b) Give a function $g(n)$ so that the number of syllables in the pronunciation of n is $\Omega(g(n))$ for all $n \geq 1$.
 - (c) Give a function $h(n)$ so that the number of syllables in the pronunciation of n is $\Theta(h(n))$ for all $n \geq 1$.
3. (3 points) Prove that $\log_a n = \Theta(\log_b n)$ for any positive constants $a \neq 1, b$ by proving each of the following. You must use the definition of big-O and big-Omega in your proofs; that is, you must give a c and an n_0 . (Hint: Use the change of base formula for logarithms, which states that $\log_a n = \frac{\log_b n}{\log_b a}$.)
 - (a) $\log_a n = O(\log_b n)$
 - (b) $\log_a n = \Omega(\log_b n)$
4. (4 points) For each of the following algorithms, you will analyze both the worst and best case runtimes.

(a)

IsPrime(n):
if $n < 2$
return False
for $d = 2$ to $\lfloor \sqrt{n} \rfloor$
if $n \bmod d = 0$
return False
return True

- i. Describe a best-case input to IsPrime.
- ii. Give a function $f(n)$ so that the best-case runtime of IsPrime is $\Theta(f(n))$.
- iii. Describe a worst-case input to IsPrime.
- iv. Give a function $f(n)$ so that the worst-case runtime of IsPrime is $\Theta(f(n))$.

(b)

HasDuplicate(A[1..n]):
for $i = 1$ to n
for $j = i + 1$ to n
if $A[i] = A[j]$
return True
return False

- i. Describe a best-case input to HasDuplicate.
- ii. Give a function $f(n)$ so that the best-case runtime of HasDuplicate is $\Theta(f(n))$.
- iii. Describe a worst-case input to HasDuplicate.
- iv. Give a function $f(n)$ so that the worst-case runtime of HasDuplicate is $\Theta(f(n))$.